

Mrs Cooper
Mr Feng
Mrs Millar
Mrs Squires

Ms Davis
Mrs Greenberg
Ms Smith
Mrs Ward

Name:

Teacher:



Pymble Ladies' College

YEAR 11 YEARLY EXAMINATION 2022

Mathematics Advanced

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using non-erasable black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning or calculations

**Total marks
72****Section 1 – 10 marks** (pages 1-3)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 62 marks (pages 6-19)

- Attempt Questions 11-28
- Allow about 1 hour and 45 minutes for this section

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SECTION I

10 Marks

Use the Multiple-Choice Answer sheet for Questions 1-10

1 What is the equation of the line perpendicular to the line with equation $3y = -2x + 1$?

(A) -2

(B) $-\frac{2}{3}$

(C) $\frac{1}{2}$

(D) $\frac{3}{2}$

2 Which of the following is a one-to-one function?

(A) $y = x^3 - x$

(B) $y = x^2 + 4$

(C) $y = 1 - x^3$

(D) $y = \sqrt{9 - x^2}$

3 What are the radius and centre of a circle with equation $x^2 - 6x + y^2 + 10y + 18 = 0$?

(A) Radius $= 3\sqrt{2}$, Centre $(-3, 5)$

(B) Radius $= 3\sqrt{2}$, Centre $(3, -5)$

(C) Radius $= 4$, Centre $(-3, 5)$

(D) Radius $= 4$, Centre $(3, -5)$

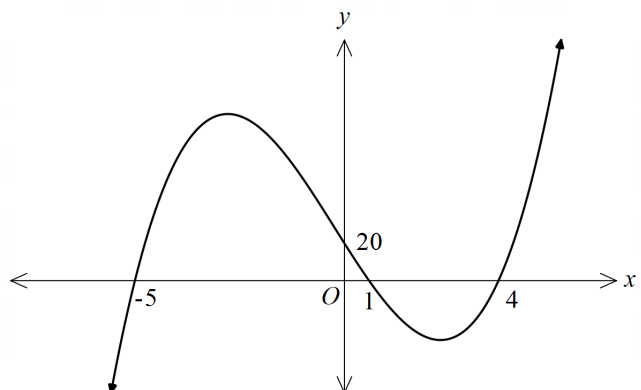
4 The sketch of $y = f(x)$ is shown.
Which of the following is **NOT** a solution to $f(x) = 0$?

(A) $x = -5$

(B) $x = 1$

(C) $x = 4$

(D) $y = 20$

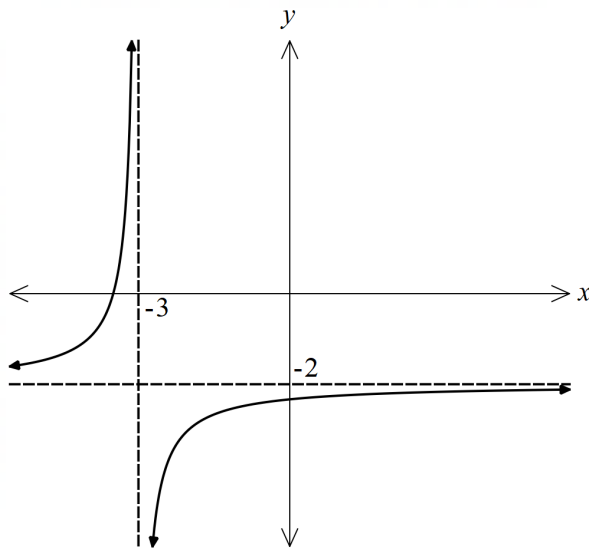


- 5 If $\tan \theta = \frac{2}{3}$ and θ is acute, which of the following is the exact value of $\cos \theta$?
- (A) $\frac{2}{\sqrt{5}}$
- (B) $\frac{3}{\sqrt{5}}$
- (C) $\frac{3}{\sqrt{13}}$
- (D) $\frac{2}{\sqrt{13}}$
- 6 What is the derivative of $(x^3 + 4)^3$?
- (A) $3(x^3 + 4)^2$
- (B) $3(3x^2 + 4)^2$
- (C) $9x^2(x^3 + 4)^2$
- (D) $9x^2(3x^2 + 4)^2$
- 7 Which of the following is an expression for $\cos^2 x + \cot^2 x + \sin^2 x$?
- (A) 2
- (B) $\sec^2 x$
- (C) $\operatorname{cosec}^2 x$
- (D) $1 + \operatorname{cosec}^2 x$
- 8 Given $\log_3 5 = 1.46$ and $\log_3 2 = 0.63$, evaluate $\log_3 100$.
- (A) 2.09
- (B) 2.72
- (C) 3.55
- (D) 4.18

9 Let $f(x) = \frac{1}{x}$ and $g(x) = x + 2$. What is the domain of $f(g(x))$?

- (A) All real x
- (B) All real x , $x \neq -2$
- (C) All real x , $x \neq 0$
- (D) All real x , $x \neq 2$

10 The graph of $y = \frac{-1}{x+b} + c$ is shown.



The values for b and c are:

- (A) $b = 3$ $c = 2$
- (B) $b = 3$ $c = -2$
- (C) $b = -3$ $c = 2$
- (D) $b = -3$ $c = -2$

End of Section I

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Name:

Teacher:



Pymble Ladies' College

**YEAR 11
YEARLY EXAMINATION
2022**

**Mathematics Advanced
Section II**

62 marks

Attempt Questions 11-28

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Additional writing space is provided at the end of the paper.
- Write using black or blue pen. Black is preferred.
- Erasable pens are **not** to be used.
- Your responses should include relevant mathematical reasoning and/or calculations.

SECTION II

62 marks

Attempt all questions

Name:.....

Teacher's Name:.....

Answer in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

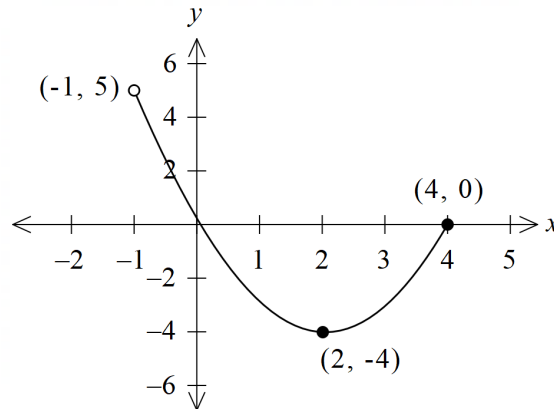
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Question 11

Marks

The graph of the function $f(x)$ is shown below.

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Using interval notation, state the domain and range of $f(x)$.

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Question 12

Marks

$(3 + \sqrt{a})(4 + \sqrt{a}) = 17 + k\sqrt{a}$ where a and k are positive integers.

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Find the values of a and k .

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Question 13**Marks**Evaluate $\log_3 27$.**1**

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Question 14**Marks**Find the equation of the tangent to the curve $y = x^2 + 3x$ at the point $(1, 4)$.**3**

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Question 15**Marks**

Find the derivative of $f(x) = 5x^2 - 2x$ by first principles.

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Question 16**Marks**

Show working to determine whether the function $f(x) = \frac{x^3}{x^2 - 1}$ is odd, even or neither.

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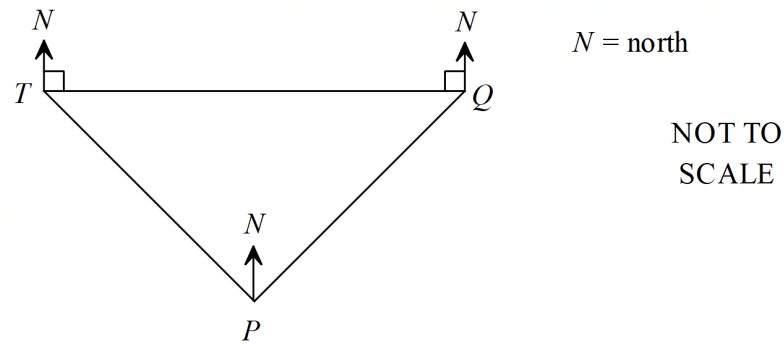
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Question 17

Marks



The diagram represents the flight path of a plane. the arrows marked with N in the diagram indicate the direction of north.

The plane flies from a town, T , to a property, P , which is 83 kilometres away on a bearing of 129° . The plane then flies to another property, Q , which is 91 kilometres from P , and which is due east of T .

(a) Mark the diagram (above) with all the given information. **1**

(b) What is the bearing of Q from P correct to the nearest minute? **2**

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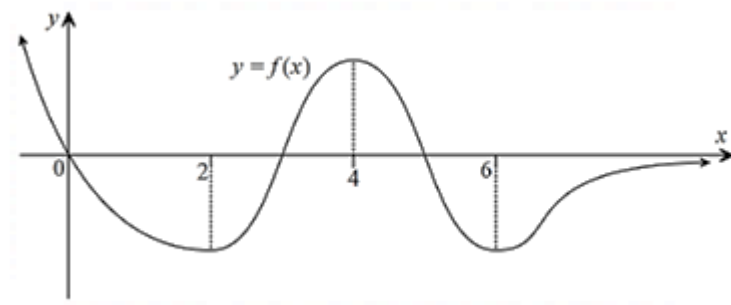
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Question 18**Marks**

Consider the function $y = f(x)$ shown below.

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For which values of x is the function increasing?

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Question 19**Marks**

Solve $2 \cos^2 x - \cos x = 0$ in the domain $[0, 2\pi]$.

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Question 20**Marks**

Consider the *We are Jeans* store. It costs the business \$29 to make a pair of jeans. This includes materials and labour. The fixed costs come to \$1000. The store sells the jeans for \$49 per pair.

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How many pairs of jeans do they have to sell to start making a profit on them?

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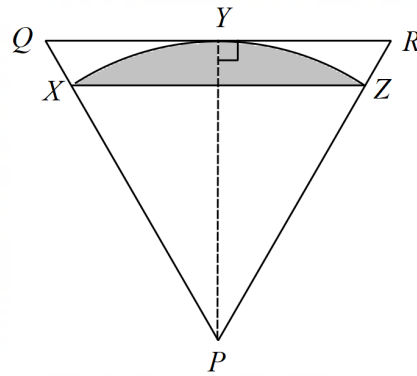
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Question 21

Marks



NOT TO
SCALE

The diagram shows an equilateral triangle PQR with sides 6 cm in length, and XYZ is an arc of a circle with centre P . The point Y lies on QR .

- (a) Show that the length of the radius $PY = 3\sqrt{3}$ cm.

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- (b) Find the length of the arc XYZ .

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- (c) Find the **EXACT** area of the shaded segment XYZ .

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Question 22**Marks**

Solve each of the following:

(a) $|1 - 5x| = 16.$

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(b) $x^2 + 24 = 11x.$

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(c) $2^{x-3} = \frac{1}{4}.$

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(d) $\log_x 4 = \log_3 \sqrt{3}.$

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Question 23**Marks**Differentiate each of the following with respect to x .

(a) $y = 7 - \frac{3}{x^2}$

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(b) $y = \frac{2x-5}{3x+7}$

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(c) $y = \frac{1}{\sqrt{5-x}}$

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Question 24**Marks**

Prove $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$.

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Question 25**Marks**

Air pressure P , measured in kPa, at an altitude of h metres above sea level can be approximated using the formula $P = 101e^{-kh}$, where k is a constant. The air pressure is 90 kPa at an altitude of 1000 m.

- (a) Find the value of k correct to 4 significant figures.

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- (b) Find the depth of a mine below sea level, where the air pressure is 103 kPa.

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Question 26**Marks**

A particle is moving in a straight line. At time t seconds its displacement, in metres, from a fixed point O on the line is given by $x = t^2 - 2t$.

(a) When is the particle at rest?

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(b) When does the particle return to its starting position?

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(c) What is the velocity of the particle at the time it has returned to its starting position?

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Question 27**Marks**

A swimming pool is to be emptied for maintenance. The quantity of water, Q litres, remaining in the pool at a time, t minutes after it starts to drain, is given by

$$Q(t) = 2000(25 - t)^2, \quad t \geq 0$$

- (a) What is the initial amount of water in the pool?

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- (b) At what rate is the water flowing out of the pool at 2 minutes?

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- (c) At what time does the rate of flow of water from the pool reach 20 000 L/minute?

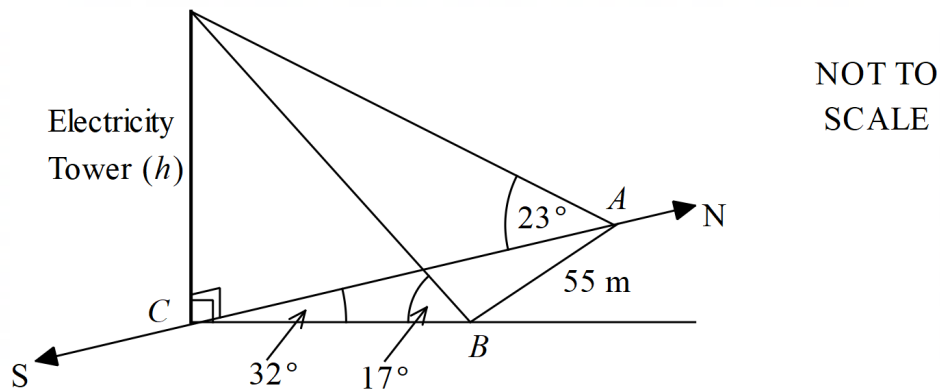
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Question 28

Marks

A person walks on level ground, in a northerly direction, away from an electricity tower. When they arrive at a point A , the angle of elevation to the top of the tower is 23° . Another person walks on level ground on a bearing of 032°T from the same tower, until he reaches point B , and notices that the angle of elevation is 17° . It is known that points A and B are 55 m apart. Let h be the height of the tower and assume that the tower, base C , is perpendicular to the ground.



- (a) Find BC in terms of h .

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- (b) Given $AC = h \cot 23^\circ$, show that $h^2 = \frac{3025}{\cot^2 23^\circ + \cot^2 17^\circ - 2 \cot 23^\circ \cot 17^\circ \cos 32^\circ}$.

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Question 28 continues on page 19

Question 28 continued.

(c) Find h correct to the nearest metre.

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END OF PAPER

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NSW Education Standards Authority

2021 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

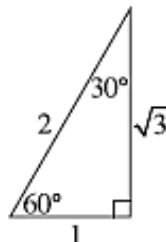
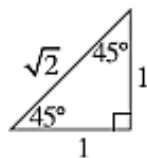
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

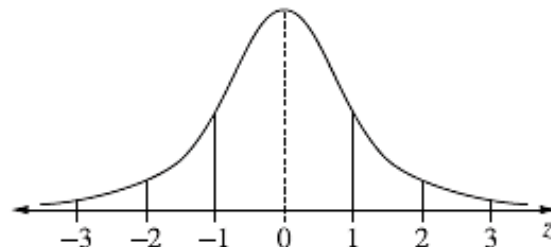
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$= \frac{b-a}{2n} \{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

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Name: *Answers*

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Mathematics Advanced

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**Total marks
72**

Section 1 – 10 marks (pages 1-3)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 62 marks (pages 6-19)

- Attempt Questions 11-28
- Allow about 1 hour and 45 minutes for this section

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SECTION I

10 Marks

Use the Multiple-Choice Answer sheet for Questions 1-10

- 1 What is the ~~equation~~^{gradient} of the line perpendicular to the line with equation $3y = -2x + 1$?

(A) -2

(B) $-\frac{2}{3}$

(C) $\frac{1}{2}$

(D) $\frac{3}{2}$

- 2 Which of the following is a one-to-one function?

(A) $y = x^3 - x$

(B) $y = x^2 + 4$

(C) $y = 1 - x^3$

(D) $y = \sqrt{9 - x^2}$

- 3 What are the radius and centre of a circle with equation $x^2 - 6x + y^2 + 10y + 18 = 0$?

(A) Radius = $3\sqrt{2}$, Centre $(-3, 5)$

(B) Radius = $3\sqrt{2}$, Centre $(3, -5)$

(C) Radius = 4, Centre $(-3, 5)$

(D) Radius = 4, Centre $(3, -5)$

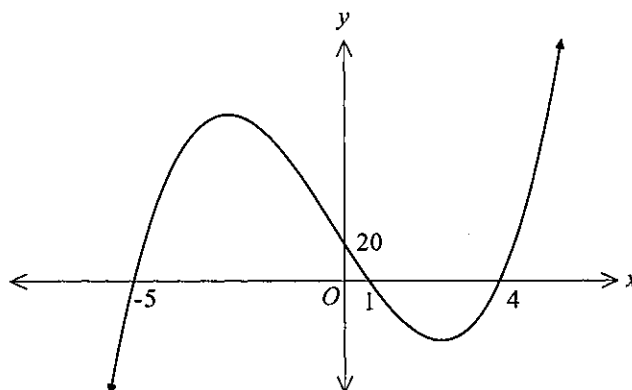
- 4 The sketch of $y = f(x)$ is shown.
Which of the following is **NOT** a solution to $f(x) = 0$?

(A) $x = -5$

(B) $x = 1$

(C) $x = 4$

(D) $y = 20$



5 If $\tan \theta = \frac{2}{3}$ and θ is acute, which of the following is the exact value of $\cos \theta$?

(A) $\frac{2}{\sqrt{5}}$

(B) $\frac{3}{\sqrt{5}}$

(C) $\frac{3}{\sqrt{13}}$

(D) $\frac{2}{\sqrt{13}}$

6 What is the derivative of $(x^3 + 4)^3$?

(A) $3(x^3 + 4)^2$

(B) $3(3x^2 + 4)^2$

(C) $9x^2(x^3 + 4)^2$

(D) $9x^2(3x^2 + 4)^2$

7 Which of the following is an expression for $\cos^2 x + \cot^2 x + \sin^2 x$?

(A) 2

(B) $\sec^2 x$

(C) $\operatorname{cosec}^2 x$

(D) $1 + \operatorname{cosec}^2 x$

8 Given $\log_3 5 = 1.46$ and $\log_3 2 = 0.63$, evaluate $\log_3 100$.

(A) 2.09

(B) 2.72

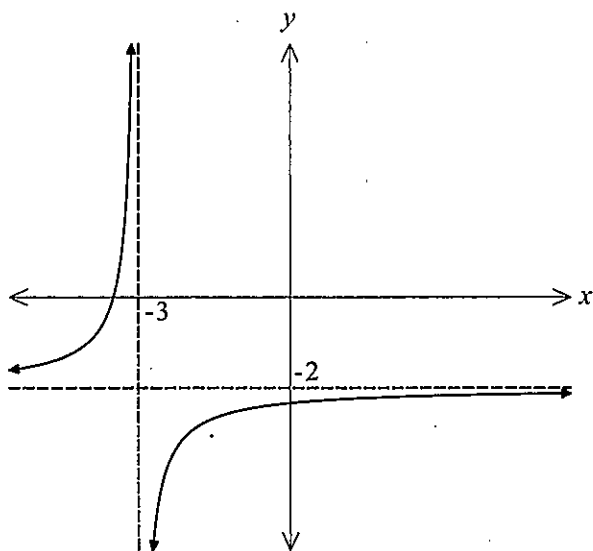
(C) 3.55

(D) 4.18

9 Let $f(x) = \frac{1}{x}$ and $g(x) = x + 2$. What is the domain of $f(g(x))$?

- (A) All real x
- ☒ (B) All real x , $x \neq -2$
- (C) All real x , $x \neq 0$
- (D) All real x , $x \neq 2$

10 The graph of $y = \frac{-1}{x+b} + c$ is shown.



The values for b and c are:

- (A) $b = 3$ $c = 2$
- ☒ (B) $b = 3$ $c = -2$
- (C) $b = -3$ $c = 2$
- (D) $b = -3$ $c = -2$

End of Section I

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Name:

Teacher:



Pymble Ladies' College

**YEAR 11
YEARLY EXAMINATION
2022**

**Mathematics Advanced
Section II**

62 marks

Attempt Questions 11-28

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Additional writing space is provided at the end of the paper.
- Write using black or blue pen. Black is preferred.
- Erasable pens are **not** to be used.
- Your responses should include relevant mathematical reasoning and/or calculations.

SECTION II

62 marks

Attempt all questions

Name:.....

Teacher's Name:.....

Answer in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

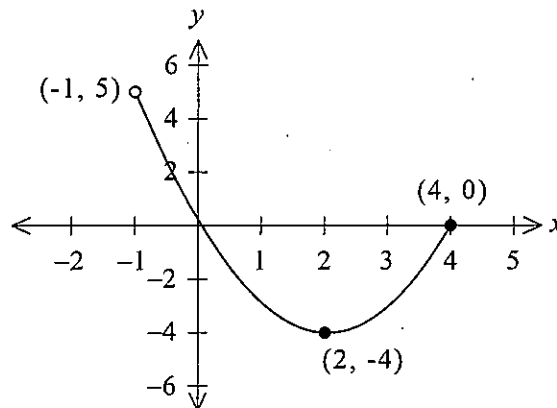
Extra writing space is provided at the end of this paper.

Question 11

Marks

The graph of the function $f(x)$ is shown below.

2



Using interval notation, state the domain and range of $f(x)$.

Domain $[-1, 4]$ Range $[-4, 5)$

1 for each

IF not interval notation but correct, -1

Question 12

Marks

$(3 + \sqrt{a})(4 + \sqrt{a}) = 17 + k\sqrt{a}$ where a and k are positive integers.

2

Find the values of a and k .

$$12 + 7\sqrt{a} + a = 17 + k\sqrt{a} \quad *$$

$$\therefore 12 + a = 17 \rightarrow a = 5$$

$$k = 7$$

2 | Correct.
1 | Getting to *

Question 13

Marks

Evaluate $\log_3 27$.

1

$$\log_3 3^3 = 3 \log_3 3 \quad \text{1 r/w}$$

$$= 3$$

Question 14

Marks

Find the equation of the tangent to the curve $y = x^2 + 3x$ at the point $(1, 4)$.

3

$$\frac{dy}{dx} = 2x + 3 \quad \text{When } x = 1 \quad \frac{dy}{dx} = 2 + 3$$

$$= 5$$

$$\text{Eqn of tangent: } y - 4 = 5(x - 1)$$

$$y = 5x - 1$$

3	Correct
2	Derivative, substitution but error in finding tangent
1	Derivative and substitution

Question 15

Marks

Find the derivative of $f(x) = 5x^2 - 2x$ by first principles.

3

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{5(x+h)^2 - 2(x+h) - (5x^2 - 2x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 2x - 2h - 5x^2 + 2x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(10x + 5h - 2)}{h} \\
 &= \lim_{h \rightarrow 0} 10x + 5h - 2 \\
 &= 10x - 2
 \end{aligned}$$

3	correct working and answer
2	correct algebraic expansion and attempts to apply lim as $h \rightarrow 0$
1	<u>correct</u> formula stated.

Question 16

Marks

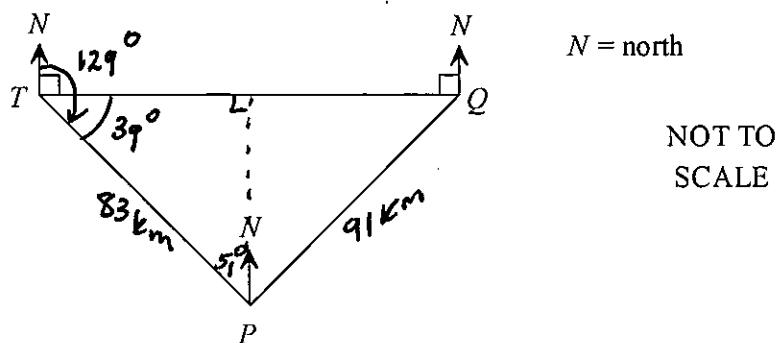
Show working to determine whether the function $f(x) = \frac{x^3}{x^2 - 1}$ is odd, even or neither.

2

$$\begin{aligned}
 f(a) &= \frac{a^3}{a^2 - 1} \\
 f(-a) &= \frac{(-a)^3}{(-a)^2 - 1} \\
 &= \frac{-a^3}{a^2 - 1} \\
 &= -f(a)
 \end{aligned}$$

since $f(-a) = -f(a)$
the function is odd

2	reaches $f(-x) = -f(x)$ correctly
1	no conclusion, no mark
1	substitutes correctly with brackets for $f(-x)$



The diagram represents the flight path of a plane. the arrows marked with N in the diagram indicate the direction of north.

The plane flies from a town, T , to a property, P , which is 83 kilometres away on a bearing of 129° . The plane then flies to another property, Q , which is 91 kilometres from P , and which is due east of T .

- (a) Mark the diagram (above) with all the given information.

1

- (b) What is the bearing of Q from P correct to the nearest minute?

2

$$\begin{aligned} \text{In } \triangle TPQ \quad \frac{\sin Q}{83} &= \frac{\sin 39^\circ}{91} \\ \sin Q &= \frac{83 \sin 39^\circ}{91} \\ Q &= 35^\circ 1' 46'' \text{ or } 144^\circ 58' 14'' \\ &\quad \text{Not possible} \\ \therefore Q &= 35^\circ 1' 46'' \\ \therefore \angle TPQ &= 180 - (39 + 35^\circ 1' 46'') \\ &= 105^\circ 58' 14'' \end{aligned}$$

$$\begin{aligned} \text{Bearing of } Q \text{ from } P &: 105^\circ 58' 14'' - 51^\circ \\ &= 054^\circ 58' \text{ (nearest minute)} \end{aligned}$$

2 | Correct answer $054^\circ 58'$

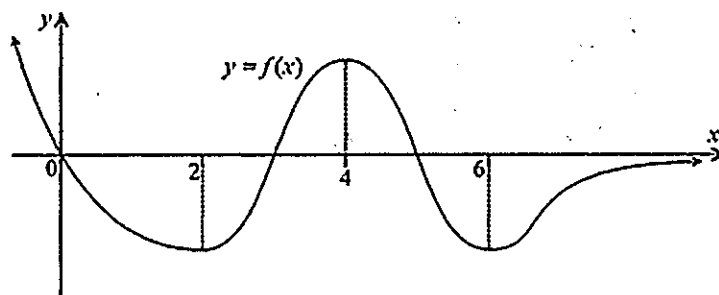
1 | Works towards
 $\angle Q = 35^\circ 1' 46''$ using
 Sine Rule (or other
 correct method)

Question 18

Marks

Consider the function $y = f(x)$ shown below.

2



For which values of x is the function increasing?

1 for each.

$$2 < x < 4 \text{ or } x > 6$$

OR $x \in (2, 4) \cup (6, \infty)$

2	correct numbers
1	correct parentheses

Question 19

Marks

Solve $2 \cos^2 x - \cos x = 0$ in the domain $[0, 2\pi]$.

3

$$\cos x (2 \cos x - 1) = 0$$

$$\therefore \cos x = 0 \text{ or } \cos x = \frac{1}{2}$$

$$x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, 2\pi \text{ or } x = \frac{\pi}{3}, \frac{5\pi}{3}$$

3	4 correct angles
---	------------------

2	2 correct angles
---	------------------

1	Factorises correctly
---	----------------------

Question 20

Marks

Consider the *We are Jeans* store. It costs the business \$29 to make a pair of jeans. This includes materials and labour. The fixed costs come to \$1000. The store sells the jeans for \$49 per pair.

3

How many pairs of jeans do they have to sell to start making a profit on them?

Let the number of jeans be n

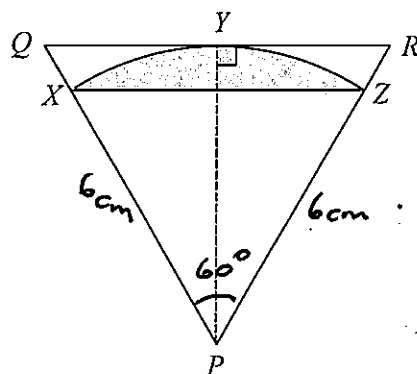
$$49n = 29n + 1000$$

$$20n = 1000$$

$$n = 50$$

They need to sell more than 50 pairs to begin to make a profit.

3	$n > 50$
2	$n = 50$
1	Works towards $20n = 1000$

NOT TO
SCALE

The diagram shows an equilateral triangle PQR with sides 6 cm in length, and XYZ is an arc of a circle with centre P . The point Y lies on QR .

- (a) Show that the length of the radius $PY = 3\sqrt{3}$ cm.

1

$$\begin{aligned} \text{In } \triangle PYQ \quad \cos 30^\circ &= \frac{PY}{6} & 1 \text{ r/w} \\ PY &= 6 \times \frac{\sqrt{3}}{2} \\ &= 3\sqrt{3} \text{ cm} \end{aligned}$$

- (b) Find the length of the arc XYZ .

1

$$\begin{aligned} \angle QPR &= \frac{\pi}{3} & l_{XYZ} &= 3\sqrt{3} \times \frac{\pi}{3} & 1 \text{ r/w} \\ & & &= \sqrt{3} \pi \text{ cm} \end{aligned}$$

- (c) Find the **EXACT** area of the shaded segment XYZ .

2

$$\begin{aligned} \text{Shaded area} &= \text{Area of sector} - \text{Area } \triangle XPZ \\ &= \frac{1}{2} \times (3\sqrt{3})^2 \times \frac{\pi}{3} - \frac{1}{2} \times (3\sqrt{3})^2 \sin \frac{\pi}{3} \\ &= \frac{9\pi}{2} - \frac{27 \times \sqrt{3}}{2} \\ &= \left(\frac{9\pi}{2} - \frac{27\sqrt{3}}{4} \right) \text{ cm}^2 \end{aligned}$$

2 | Correct
answer

1 | $A(\text{sector}) - A(\triangle XPZ)$
OR
uses $A = \frac{1}{2} r^2 (\theta - \sin \theta)$
(with errors)

Question 22

Marks

Solve each of the following:

(a) $|1-5x|=16$.

2

$1-5x = \pm 16$			
$1-5x = 16$	or	$1-5x = -16$	2 both correct
$5x = -15$		$-5x = -17$	
$x = -3$		$x = 17/5$	1 one correct

(b) $x^2+24=11x$.

$x^2-11x+24=0$		2 correct solutions
$(x-8)(x-3)=0$		
$\therefore x=3, 8$		1 correct factorisation

(c) $2^{x-3}=\frac{1}{4}$.

2

$2^{x-3} = 2^{-2}$		2 correct answer
$\therefore x-3 = -2$		
$x = 1$		1 progress to equate indices OR correctly uses identity

(d) $\log_x 4 = \log_3 \sqrt{3}$.

2

$\log_x 4 = \frac{1}{2} \log_3 3$	$x = 16$	2 correct answer
$\log_x 4 = \frac{1}{2}$		
$x^{1/2} = 4$		1 progress using identity

Question 23

Marks

Differentiate each of the following with respect to x .

(a) $y = 7 - \frac{3}{x^2}$

$y = 7 - 3x^{-2}$

2

$\frac{dy}{dx} = \frac{6}{x^3}$

2 | Correct differentiation

OR $\frac{dy}{dx} = 6x^{-3}$

1 | Rewrite as $y = 7 - 3x^{-2}$

(b) $y = \frac{2x-5}{3x+7}$

2

$\frac{dy}{dx} = \frac{(3x+7)(2) - (2x-5)(3)}{(3x+7)^2}$

$= \frac{6x+14-6x+15}{(3x+7)^2}$

2 | Correct simplified answer

1 | Correct numerator OR correct denominator

$= \frac{29}{(3x+7)^2}$

(c) $y = \frac{1}{\sqrt{5-x}}$

$y = (5-x)^{-1/2}$

2

$\frac{dy}{dx} = -\frac{1}{2}(5-x)^{-3/2} \times -1$

2 | correct simplified answer

$= \frac{1}{2\sqrt{(5-x)^3}}$

1 | Correct coefficient $(-\frac{1}{2})$ and index $(-\frac{3}{2})$ OR

correctly multiplies by chain rule factor

Question 24

Marks

Prove $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} = 2\sec^2\theta$.

2

LHS : $\frac{1-\sin\theta + 1+\sin\theta}{(1+\sin\theta)(1-\sin\theta)}$

$= \frac{2}{1-\sin^2\theta}$ $1-\sin^2\theta = \cos^2\theta$

$= \frac{2}{\cos^2\theta}$

$= 2\sec^2\theta$

$= \text{RHS}$

2 | fully complete proof

1 | combines or breaks apart fractions (showing numerator) using $(1+\sin\theta)(1-\sin\theta) = 1-\sin^2\theta$

OR

correctly uses $\frac{1}{\cos^2\theta} = \sec^2\theta$

Question 25

Marks

Air pressure P , measured in kPa, at an altitude of h metres above sea level can be approximated using the formula $P = 101e^{-kh}$, where k is a constant. The air pressure is 90 kPa at an altitude of 1000 m.

- (a) Find the value of k correct to 4 significant figures.

2

$90 = 101e^{-1000k}$

$e^{-1000k} = \frac{90}{101}$

$e^{-1000k} = 0.891$

$k = -\frac{1}{1000} \ln(0.891)$

$k = 0.001153$

2 | Correctly solves using logs.

1 | correct substitution and rearrange to index form

- (b) Find the depth of a mine below sea level, where the air pressure is 103 kPa.

3

$10.3 = 10.1e^{-kh}$ $h = 17.00488027$

$= 17 \text{ m.}$

$e^{-kh} = \frac{103}{101}$

$-kh = \ln\left(\frac{103}{101}\right)$

$h = -\frac{1}{k} \ln\left(\frac{103}{101}\right)$

3 | Correctly solves for $h = -170$ or answers in context (170 m BELOW)

2 | Writes a correct equation using logs.

1 | substitutes $P=103$ and correctly rounds value of k .

Question 26

Marks

A particle is moving in a straight line. At time t seconds its displacement, in metres, from a fixed point O on the line is given by $x = t^2 - 2t$.

- (a) When is the particle at rest?

2

$$\frac{dx}{dt} = 2t - 2 \quad 2t - 2 = 0$$

$$t = 1$$

When $\frac{dx}{dt} = 0 \quad \therefore$ particle is at rest when $t = 1$ s.

2	Correct answer.
1	Correct $\frac{dx}{dt}$

- (b) When does the particle return to its starting position?

1

When $t = 0 \quad x = 0 \quad \therefore$ starting position is $x = 0$

$$t^2 - 2t = 0$$

$$t(t - 2) = 0 \quad \therefore$$

$\therefore t = 0$ or $t = 2$ to starting position after 2 s

1/r/w

- (c) What is the velocity of the particle at the time it has returned to its starting position?

1

When $t = 2 \quad \frac{dx}{dt} = 2 \times 2 - 2$

$$\frac{dx}{dt} = 2 \text{ m/s}$$

1 r/w

Question 27

Marks

A swimming pool is to be emptied for maintenance. The quantity of water, Q litres, remaining in the pool at a time, t minutes after it starts to drain, is given by

$$Q(t) = 2000(25-t)^2, \quad t \geq 0$$

- (a) What is the initial amount of water in the pool?

1

When $t = 0$ $Q(t) = 2000 \times 25^2$ 1 r/w
 $= 1250000 \text{ litres}$

- (b) At what rate is the water flowing out of the pool at 2 minutes?

2

$\frac{dQ}{dt} = 2000 \times 2(25-t) \times -1$
 $= -4000(25-t)$ 2 | correct answer.

When $t = 2$ $\frac{dQ}{dt} = -4000(25-2)$ 1 | correct $\frac{dQ}{dt}$
 $= -92000 \text{ L/min}$

- (c) At what time does the rate of flow of water from the pool reach 20 000 L/minute?

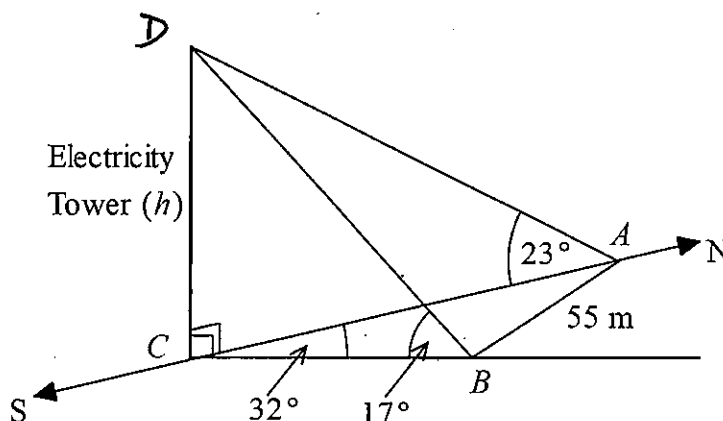
2

When $\frac{dQ}{dt} = -20000$ 2 | Substituting -20000 getting 20 min
 $-20000 = -4000(25-t)$
 $25-t = 5$
 $t = 20 \text{ min}$ 1 | Substituting 20000, giving 30 min

Question 28

Marks

A person walks on level ground, in a northerly direction, away from an electricity tower. When they arrive at a point A , the angle of elevation to the top of the tower is 23° . Another person walks on level ground on a bearing of 032°T from the same tower, until he reaches point B , and notices that the angle of elevation is 17° . It is known that points A and B are 55 m apart. Let h be the height of the tower and assume that the tower, base C , is perpendicular to the ground.



NOT TO
SCALE

- (a) Find BC in terms of h .

1

In $\triangle BCD$ $\tan 17^\circ = \frac{h}{BC}$ 1 r/w

$BC = h \cot 17^\circ$

- (b) Given $AC = h \cot 23^\circ$, show that $h^2 = \frac{3025}{\cot^2 23^\circ + \cot^2 17^\circ - 2 \cot 23^\circ \cot 17^\circ \cos 32^\circ}$.

2

In $\triangle ABC$

$55^2 = (h \cot 17^\circ)^2 + (h \cot 23^\circ)^2 - 2(h \cot 17^\circ)(h \cot 23^\circ) \cos 32^\circ$

$3025 = h^2(\cot^2 17^\circ + \cot^2 23^\circ - 2 \cot 17^\circ \cot 23^\circ \cos 32^\circ)$

$h^2 = \frac{3025}{(\cot^2 17^\circ + \cot^2 23^\circ - 2 \cot 17^\circ \cot 23^\circ \cos 32^\circ)}$

As req'd

Question 28 continues on page 19

2	Answer
1	X this line. including h throughout on the RHS.

Question 28 continued.

(c) Find h correct to the nearest metre.

1

$$\begin{aligned} h &= 30.84731622 && 1 \text{ r/w} \\ &\approx 31 \text{ m} \end{aligned}$$

END OF PAPER